

## Chapter 2

# Adaptive dynamics; an asymptotic point of view

So far, we have given examples of physiologically structured populations, i.e., structured by a parameter describing a biological, physiological or ecological characteristic of the individuals. Further examples are also described in Chapter 4 below. When this characteristic is inherent to the individual, i.e., it is fixed at the very beginning of its life, we refer to it as a *trait*; we prefer to avoid calling it a phenotype. The theory which focusses on phenotypic evolution driven by *small mutations* in replication, while ignoring both sex and genes, is known by the name *Adaptive dynamics* and is part of *Evolution theory*, see [172, 115, 80, 81, 169, 45] and the references given therein. A general mathematical treatment of the general subject of selection vs mutation can also be found in [43] (in particular population geneticists might prefer the assumption that mutations are rare rather than small).

The two main ingredients in this theory are (i) the selection principle which favors the population with best adapted trait, and (ii) mutations which allow off-springs to have slightly different traits than their mother. The combination of the two effects is studied by adaptive dynamics. This turns out to be an extremely intricate theory on which several possible mathematical approaches are possible. One of the reasons is that it is merely impossible to consider this problem without introducing small parameters (mutations can be small or rare for instance, population should be large in any case but relative death rates can vary). Therefore *adaptive evolution* can be studied with various mathematical tools. Evolutionary game theory is a standard point of view, see for instance [135, 136, 134] after it was introduced by J. Maynard Smith (see [169]). Probability theory is also natural because fluctuations are important at the individual level and individual centered models are naturally stochastic. Departing from an individual centered stochastic dynamics, several possible limits are possible as the population becomes

large ([101]). Here we look at models which are in one special class of those limits.

In this chapter, we give a first and very elementary point of view based on structuring an ODE and including mutations. Our main goal is to apply the corresponding asymptotic theory and show how the concept of monomorphic population arises naturally in the limit of small mutations over a long time compared to one generation length. Such rescaling has also been used for the spread of genetic traits, in a probabilistic framework (see [110] and the references therein).

Our presentation in this chapter follows the lines of [83] but uses a simpler framework for the population model. We begin with several simple examples of physiologically structured population (without mutations) where a selection principle can be proved. Then we introduce the mutations and this raises the question of finding the appropriate scales. Here, and this is one of the possible scales, we assume that mutations are frequent but have a very small effect on the trait. This allows us to state an asymptotic problem. We show that it leads to the selection of a single trait (monomorphic population) but the dynamics of this trait is far from obvious. A differential equation is not enough to describe it and we introduce, following [83], a Hamilton–Jacobi equation. The possible asymptotic behaviors of this trait (monomorphic or dimorphic) are obtained through numerical simulations: continuous evolution, jump, branching from a monomorphic to a dimorphic population or junction from dimorphic to monomorphic situations.

Our motivation for the model we use in this chapter comes from the chemostat equation (1.30). Consider that the time scale for food consumption is faster than the time scale of birth, then one naturally ends up with an algebraic equation for the substrate that is the system

$$\begin{cases} S(t) = S_0 - \int \eta(x, S(t))n(t, x)dx, & t \geq 0, x > 0 \\ \frac{\partial}{\partial t}n(t, x) = n(t, x)(\eta(x, S) - R), \\ n(t = 0, x) = n^0(x) > 0, n^0 \in L^1 \cap L^\infty(\mathbb{R}^+). \end{cases} \quad (2.1)$$

When  $\eta$  is linear in  $S$ , we arrive at the more explicit formula

$$S(t) = Q_b \left( \int \eta(x)n(t, x)dx \right)$$

and this is the form we take in this chapter as a compromise between generality and simplicity.

## 2.1 Structured population and selection principle: a simple case

The first question we consider here is how to give a mathematical description of the process in which some specific trait is selected in a given environment. It is the best

adapted trait in terms of using resources and that trait is called an *Evolutionary Stable Strategy* (ESS in short). The origin of this denomination ([170]) comes from evolution theory; no mutant with a different trait can invade a population with the trait corresponding to an ESS. An example of the selection principle has already been mentioned, in the case of the chemostat, see Lemma 1.2. Here we give an easy example that can be treated by explicit computations. In Section 2.2 we give a more general framework.

To begin with a simple example, we look at a variant of Verhulst's logistic equation (see Section 1.1) which we structure with a trait  $x \in \mathbb{R}$  and we illustrate the selection principle on this very simple example. We consider that the reproduction rate depends on the trait, i.e.,  $b = b(x) > 0$ , and that the death rate is proportional to the total population number (this is what happens in logistic growth, see Section 1.1). We arrive at

$$\begin{cases} \frac{d}{dt}n(t, x) = b(x)n(t, x) - \varrho(t)n(t, x), \\ \varrho(t) = \int_{\mathbb{R}} n(t, x)dx, \\ n(t=0) = n^0(x) > 0 \text{ for } x \in ]x_m, x_M[, \quad n^0(x) = 0 \text{ for } x \notin ]x_m, x_M[. \end{cases} \quad (2.2)$$

In this situation one can guess that the best adapted trait (highest reproduction rate here) will be selected. Indeed we have the following selection principle.

**Theorem 2.1.** *Assume that  $b \in C(\mathbb{R})$ ,  $b(\cdot) \geq \underline{b} > 0$ ,  $n^0 > 0$  and*

$$b(\bar{x}) = \max_{x \in [x_m, x_M]} b(x) \quad \text{is attained for a single } \bar{x} \in ]x_m, x_M[.$$

*Then, the solution to (2.2) satisfies*

$$\varrho(t) \rightarrow \bar{\varrho} = b(\bar{x}), \quad n(t, x) \rightarrow b(\bar{x})\delta(x - \bar{x}) \quad \text{as } t \rightarrow \infty. \quad (2.3)$$

Notice that the equation (2.2) admits many steady states, namely  $n(x) = \varrho \delta(x - y)$ ,  $\varrho = b(y)$  for any  $y$ , Theorem 2.1 selects the stable trait, the ESS. One should understand it as the trait that realizes

$$\max_x [b(x) - \bar{\varrho}] = b(\bar{x}) - \bar{\varrho} = 0.$$

The weak limit in (2.3) also indicates that the natural setting for structured population models should differ from that for differential equations because functional spaces (measures here) appear to play a role.

*Proof.* We give a proof that relies on a simple computation, another proof is possible based on more abstract arguments and that apply to more general equations (see Section 2.2). We define

$$N(t, x) = n(t, x)e^{\int_0^t \varrho(s)ds}. \quad (2.4)$$

This satisfies

$$\frac{dN(t, x)}{dt} = b(x)N(t, x),$$

and thus  $N(t, x) = n^0(x)e^{b(x)t}$ . We deduce from (2.4) that

$$\frac{d}{dt}e^{\int_0^t \varrho(s)ds} = \varrho(t)e^{\int_0^t \varrho(s)ds} = \int_{\mathbb{R}} N(t, x)dx = \int_{\mathbb{R}} n^0(x)e^{b(x)t}dx.$$

Therefore

$$\begin{aligned} e^{\int_0^t \varrho(s)ds} &= \int_{\mathbb{R}} \frac{n^0(x)}{b(x)} e^{b(x)t} dx + K, \quad K = 1 - \int_{\mathbb{R}} \frac{n^0(x)}{b(x)} dx, \\ \int_0^t \varrho(s)ds &= \ln\left(\int_{\mathbb{R}} \frac{n^0(x)}{b(x)} e^{b(x)t} dx + K\right), \\ \varrho(t) &= \int_{\mathbb{R}} n^0(x)e^{b(x)t} dx / \left[\int_{\mathbb{R}} \frac{n^0(x)}{b(x)} e^{b(x)t} dx + K\right], \end{aligned} \tag{2.5}$$

and we notice that the constant  $K$  may be negative but the denominator is larger than 1. This is a Laplace type formula and we can analyze it as follows. As  $t \rightarrow \infty$ , we have

$$\varrho(t) \leq b(\bar{x}) \int_{\mathbb{R}} \frac{n^0(x)}{b(x)} e^{b(x)t} dx / \left[\int_{\mathbb{R}} \frac{n^0(x)}{b(x)} e^{b(x)t} dx + K\right] \rightarrow b(\bar{x}),$$

because  $\int_{\mathbb{R}} \frac{n^0(x)}{b(x)} e^{b(x)t} dx \rightarrow \infty$ . For the reverse inequality, we fix an  $\varepsilon > 0$  and define the set

$$I(\varepsilon) = \{x; b(x) \geq b(\bar{x}) - \varepsilon\}.$$

Then, we have

$$\begin{aligned} \varrho(t) &\geq \int_{I(\varepsilon)} n^0(x)e^{b(x)t} dx / \left[\int_{\mathbb{R}} \frac{n^0(x)}{b(x)} e^{b(x)t} dx + K\right] \\ &\geq (b(\bar{x}) - \varepsilon) \int_{I(\varepsilon)} \frac{n^0(x)}{b(x)} e^{b(x)t} dx / \left[\int_{\mathbb{R}} \frac{n^0(x)}{b(x)} e^{b(x)t} dx + K\right] \\ &\geq \frac{b(\bar{x}) - \varepsilon}{A_\varepsilon(t)}, \end{aligned}$$

with

$$A_\varepsilon(t) = \frac{K}{\int_{I(\varepsilon)} \frac{n^0(x)}{b(x)} e^{b(x)t} dx} + \frac{\int_{\mathbb{R}} \frac{n^0(x)}{b(x)} e^{b(x)t} dx}{\int_{I(\varepsilon)} \frac{n^0(x)}{b(x)} e^{b(x)t} dx} \xrightarrow{t \rightarrow \infty} 1.$$

Therefore, we have indeed proved the first statement, namely that

$$\varrho(t) \rightarrow b(\bar{x}) \quad \text{as } t \rightarrow \infty.$$

Finally, from (2.4) and the expression of  $N(t, x)$ , we deduce

$$n(t, x) = n^0(x) e^{b(x)t - \int_0^t \varrho(s) ds},$$

and the claim (2.3) on  $n(t, x)$  follows from the limit of  $\varrho(t)$  combined with the same arguments for this Laplace formula. Indeed, for  $x \neq \bar{x}$ , we have  $b(x) < b(\bar{x})$  and thus

$$n(t, x) = n^0(x) e^{-\int_0^t (\varrho(s) - b(x)) ds} \rightarrow 0, \quad t \rightarrow \infty.$$

This proves the result.  $\square$

## 2.2 Structured population and selection principle: an extension

We now prove a selection principle in a more general setting which will be used later on and does not allow for explicit formulas as in the previous section, namely

$$\left\{ \begin{array}{l} \frac{d}{dt} n(t, x) = [b(x)Q_b(\varrho(t)) - d(x)Q_d(\varrho(t))] n(t, x), \quad x \in \mathbb{R}, \\ \varrho(t) = \int_{\mathbb{R}} n(t, x) dx, \\ n(t=0) = n^0(x) > 0. \end{array} \right. \quad (2.6)$$

The term  $b(x)Q_b(\varrho(t)) - d(x)Q_d(\varrho(t))$  can be interpreted as the fitness of individuals with the trait  $x$  being given the environment created by the total population. Here the notation is the same as in the previous section and we assume that  $b, d \in C(\mathbb{R})$ ,  $Q_b, Q_d \in C^1(\mathbb{R}^+)$  and there are  $b_m, b_M, d_M$  and  $d_m$  such that

$$b_M \geq b(\cdot) \geq b_m > 0, \quad d_M \geq d(\cdot) \geq d_m > 0, \quad (2.7)$$

$$\exists 0 < \varrho_m < \varrho_M \quad \text{such that} \quad \left\{ \begin{array}{l} \max_{x \in \mathbb{R}} [b(x)Q_b(\varrho_M) - d(x)Q_d(\varrho_M)] < 0, \\ \min_{x \in \mathbb{R}} [b(x)Q_b(\varrho_m) - d(x)Q_d(\varrho_m)] > 0, \end{array} \right. \quad (2.8)$$

$$\left\{ \begin{array}{l} \text{there is a single pair } (\bar{x}, \bar{\varrho}) \in \mathbb{R} \times [\varrho_m, \varrho_M] \text{ such that} \\ b(\bar{x})Q_b(\bar{\varrho}) - d(\bar{x})Q_d(\bar{\varrho}) = 0 = \max_{x \in \mathbb{R}} [b(x)Q_b(\bar{\varrho}) - d(x)Q_d(\bar{\varrho})]. \end{array} \right. \quad (2.9)$$

Notice that there exists a unique  $\bar{\varrho}$  that satisfies the equality

$$0 = \max_{x \in \mathbb{R}} [b(x)Q_b(\bar{\varrho}) - d(x)Q_d(\bar{\varrho})]$$

in (2.9). It is indeed a consequence of (2.8) and of the additional assumption

$$Q'_b(\varrho) < 0, \quad Q'_d(\varrho) > 0. \quad (2.10)$$

In (2.9) we assume additionally that the maximum is attained by a single point  $\bar{x}$ ; this is unessential but simplifies the statement below. Finally, we also need that there is  $R > 0$  such that for  $\varrho$  close enough to  $\bar{\varrho}$ , then

$$\beta_R := \max_{|x| \geq R} [b(x)Q_b(\varrho) - d(x)Q_d(\varrho)] < 0. \quad (2.11)$$

Again the selection principle applies in this case and gives

**Theorem 2.2.** *With assumptions (2.7)–(2.11), and assuming that  $n^0(x) > 0$  and  $\varrho_m \leq \varrho(t=0) \leq \varrho_M$ , the solution to (2.6) satisfies*

$$\varrho_m \leq \varrho(t) \leq \varrho_M, \quad \forall t \geq 0, \quad (2.12)$$

$$\varrho(t) \rightarrow \bar{\varrho}, \quad n(t, x) \rightarrow \bar{\varrho} \delta(x - \bar{x}), \quad \text{as } t \rightarrow \infty. \quad (2.13)$$

In other words, we recover that the ESS, the unbeatable strategy, is characterized by (2.9) as in the previous section.

*Proof. First step. A priori estimate.* We first prove (2.12) using assumption (2.8). Integrating the equation (2.6) in  $x$ , we obtain

$$\begin{aligned} \min_{x \in \mathbb{R}} [b(x)Q_b(\varrho(t)) - d(x)Q_d(\varrho(t))] \varrho(t) &\leq \frac{d}{dt} \varrho(t) \\ &\leq \max_{x \in \mathbb{R}} [b(x)Q_b(\varrho(t)) - d(x)Q_d(\varrho(t))] \varrho(t). \end{aligned}$$

Therefore, if  $\varrho(t)$  approaches  $\varrho_m$ , it is increasing and thus we have for all times  $\varrho_m < \varrho(t)$ , and similarly if  $\varrho(t)$  approaches  $\varrho_M$  it is decreasing and thus we have for all times  $\varrho(t) < \varrho_M$ .

*Second step. A Lyapunov functional.* We consider a function  $P(r)$  satisfying

$$rP'(r) + P(r) = Q(r) := \frac{Q_d(r)}{Q_b(r)}, \quad (2.14)$$

a differential equation on the interval  $[\varrho_m, \varrho_M]$  with several solutions given by  $P_0(r) + \frac{\lambda}{r}$  (but this family is in fact reduced to a single function as far as we are interested in the expression below).

Then we compute

$$\begin{aligned} &\frac{d}{dt} \int \left[ \frac{b(x)}{d(x)} - P(r) \right] n(t, x) dx \\ &= \int \left[ \frac{b(x)}{d(x)} - P(r) - \varrho(t)P'(\varrho(t)) \right] [b(x)Q_b(\varrho(t)) - d(x)Q_d(\varrho(t))] n(t, x) dx \\ &= \int \frac{1}{d(x)Q_b(\varrho(t))} [b(x)Q_b(\varrho(t)) - d(x)Q_d(\varrho(t))]^2 n(t, x) dx. \end{aligned}$$

As a consequence the bounded quantities  $\int [\frac{b(x)}{d(x)} - P(r)]n(t, x)dx$  are increasing and thus converge as  $t \rightarrow \infty$ ,

$$\int [\frac{b(x)}{d(x)} - P(r)]n(t, x)dx \xrightarrow{t \rightarrow \infty} L \in \mathbb{R}, \quad (2.15)$$

$$\int_0^\infty \int_{\mathbb{R}} [b(x)Q_b(\varrho(t)) - d(x)Q_d(\varrho(t))]^2 n(t, x)dxdt < \infty. \quad (2.16)$$

*Third step. The limit.* We now derive a limit for another quantity (we recall the definition of  $Q(r)$  in (2.14))

$$\begin{aligned} & \frac{d}{dt} \int [\frac{b(x)}{d(x)} - Q(\varrho(t))]^2 n(t, x)dx \\ &= \int [\frac{b(x)}{d(x)} - Q(\varrho(t))]^2 [b(x)Q_b(\varrho(t)) - d(x)Q_d(\varrho(t))]n(t, x)dx \\ & \quad - 2Q'(\varrho(t)) \int [\frac{b(x)}{d(x)} - Q(\varrho(t))]n(t, x)dx \\ & \quad \times \int [b(x)Q_b(\varrho(t)) - d(x)Q_d(\varrho(t))]n(t, x)dx. \end{aligned}$$

Therefore, using the  $L^\infty$  and non-extinction bound to upper bound the first term in the right-hand side, and Cauchy-Schwarz inequality for the second term, we also have

$$|\frac{d}{dt} \int [\frac{b(x)}{d(x)} - Q(\varrho(t))]^2 n(t, x)dx| \leq C \int [b(x)Q_b(\varrho(t)) - d(x)Q_d(\varrho(t))]^2 n(t, x)dx.$$

As a conclusion of this computation and step (ii), we deduce that

$$\frac{d}{dt} \int [\frac{b(x)}{d(x)} - Q(r)]^2 n(t, x)dx \in L^1(\mathbb{R}^+).$$

This is stronger than a mere BV bound which is enough to ensure that it has a limit, see Section 6.5.1 for details. Thanks to its integrability (second conclusion of step (ii)), we conclude

$$\int [\frac{b(x)}{d(x)} - Q(\varrho(t))]^2 n(t, x)dx \xrightarrow{t \rightarrow \infty} 0.$$

And again the use of Cauchy-Schwarz inequality results in

$$\int |\frac{b(x)}{d(x)} - Q(\varrho(t))|n(t, x)dx \xrightarrow{t \rightarrow \infty} 0. \quad (2.17)$$

Combined with the first conclusion of step (ii), we also arrive at

$$\int [\frac{b(x)}{d(x)} - P(\varrho(t))]n(t, x)dx = \int [\frac{b(x)}{d(x)} - Q(\varrho(t)) + [Q - P](\varrho(t))]n(t, x)dx \rightarrow L,$$

and thus  $[Q - P](\varrho(t))$   $\varrho(t)$  has a limit and finally

$$\varrho(t) \xrightarrow{t \rightarrow \infty} \varrho^*. \quad (2.18)$$

Indeed, on the one hand  $Q - P$  is not locally constant because

$$r(P - Q)' + P - Q = -rQ' < 0,$$

and on the other hand  $\varrho(t)$  is Lipschitz continuous (to see this, integrate in  $x$  the equation on  $n$ ). Notice at this point that we do not use the full assumption (2.10) but only the consequence that  $Q'$  does not vanish.

*Fourth step. Identifying  $\varrho^* = \bar{\varrho}$ .* Indeed, if  $\varrho^* > \bar{\varrho}$ , then for  $t$  large enough we have

$$\max_{x \in \mathbb{R}} [b(x)Q_b(\varrho(t)) - d(x)Q_d(\varrho(t))] < 0,$$

and the differential inequality of step (i) (upper bound) implies that there is extinction which is impossible since  $\varrho(t) \geq \rho_m$ . Also, if  $\varrho^* < \bar{\varrho}$ , then for  $t$  large enough we have

$$\max_{x \in \mathbb{R}} [b(x)Q_b(\varrho(t)) - d(x)Q_d(\varrho(t))] > 0,$$

and for those  $x$ 's where  $[b(x)Q_b(\varrho(t)) - d(x)Q_d(\varrho(t))] > 0$  we have exponential growth, i.e., blow-up for  $\varrho(t)$  which is again a contradiction.

*Fifth step. Weak limit for  $n(t, x)$ .* From (2.11), we have for  $t$  large enough

$$\frac{d}{dt} \int_{|x| > R} n(t, x) dx \leq \beta_R \int_{|x| > R} n(t, x) dx.$$

Therefore  $\sup_{t > 0} \int_{|x| > R} n(t, x) dx \rightarrow 0$  as  $R \rightarrow \infty$  (because this is true initially). This proves that the family  $(n(t, x))_{t > 0}$  is compact in the weak sense of measures. Therefore there are subsequences that converge weakly to measures  $n^*(x)$  and  $\bar{\varrho} = \int n^*(x) dx$  (we denote integration with measures as Lebesgue integrals to avoid specific notation).

*Sixth step. Conclusion.* From (2.17)–(2.18), we know that  $n^*(x)$  should be concentrated on the set of  $x$ 's such that  $b(x)Q_b(\bar{\varrho}) - d(x)Q_d(\bar{\varrho}) = 0$ . With assumption (2.9), this point is unique and thus  $n^*(x) = \bar{\varrho} \delta(x - \bar{x})$ . Therefore the full family  $n(t, x)$  converges (all subsequences have the same limit!) and the result is proved.  $\square$

## 2.3 Structured population: cannibalism

Models such as (2.6) are still not as general as one might wish. The simplified model of a chemostat in (2.1) provides us with an example. Cannibalism is another example of physiologically structured population where not only the total population counts. We borrow the model from [81] and references therein.

Here, the trait  $x \geq 0$  under consideration is the degree of cannibalism, and  $0 < \alpha \leq 1$  denotes the efficiency in offspring production from intraspecific predation, the (irrealistic) model is then written, for  $t \geq 0$ ,  $x \geq 0$ , as

$$\begin{cases} \frac{\partial}{\partial t} n(t, x) = rn(t, x) - \int_0^\infty yn(t, y)dy n(t, x) + \alpha x n(t, x) \int_0^\infty n(t, y)dy, \\ n(t=0, x) = n^0(x) > 0. \end{cases} \quad (2.19)$$

The parameter  $r$  is just the growth rate in the absence of cannibalism and one can see that cannibalism acts as a negative feedback. The growth rate of the population is indeed lower than the logistic rate  $r$  since

$$\frac{d}{dt} \int_0^\infty n(t, x) = \int_0^\infty n(t, x) \left[ r - (1 - \alpha) \int_0^\infty x n(t, x) \right] \leq r \int_0^\infty n(t, x). \quad (2.20)$$

In fact, one can go further and prove that such a population will go extinct, a behavior rather different from that of the previous sections and that can be interpreted as an ESS located at  $x = \infty$ .

**Theorem 2.3.** *Assume that the initial data in (2.19) decays in  $x$  so that*

$$\int_0^\infty n^0(x) e^{x^2} dx < \infty \quad \text{and} \quad \int_0^\infty x n^0(x) dx \geq \frac{r}{1 - \alpha},$$

*then there is a global positive solution to (2.19) and we have the time decay*

$$\int_0^\infty n(t, x) dx \quad \text{decays to 0 as } t \rightarrow \infty.$$

**Remark 2.1.** 1. Since, as we see in the proof,  $\int_0^\infty x n(t, x) dx$  remains positive, we conclude that the population is concentrated on a very large trait for large times as announced above.

2. The assumption  $\int_0^\infty x n^0(x) dx \geq \frac{r}{1 - \alpha}$  is not necessary (see the exercise below).

3. Of course one can doubt the possibility to undergo unlimited growth rates  $x$  and this leads to pose the same problem with  $0 \leq x \leq A$ . Then the method is similar and one can prove that the total population converges to  $\rho \delta(x = A)$ , in other words for this model the ESS is  $x = A$  (see exercise below).

*Proof.* We set  $m_k(t) = \int_0^\infty x^k n(t, x) dx$ .

*First step. Existence.* From (2.20), we derive the a priori estimate  $m_0(t) \leq m_0(0) e^{rt}$ . Therefore we have the upper bound

$$n(t, x) = n^0(x) e^{rt + \alpha x \int_0^t m_0(s) - \int_0^t m_1(s)} \leq n^0(x) e^{rt + \alpha x m_0(0) e^{rt} / r}.$$

This proves that, after using the Cauchy–Lipschitz Theorem, the solutions do not blow-up in finite time and thus they are global. In particular, thanks to our

assumption on the initial growth in  $x$ , the  $m_k(t)$  are finite but there is no extinction in finite time.

*Second step. A lower bound on  $m_1$ .* We have (using Cauchy-Schwarz inequality) some kind of Lyapunov functional

$$\frac{d}{dt}m_1(t) = m_1(t)(r - m_1(t)) + \alpha m_2(t)m_0(t) \geq m_1(t)(r - m_1(t)(1 - \alpha)). \quad (2.21)$$

Therefore we find that for all times

$$m_1(t) \geq \frac{r}{1 - \alpha}. \quad (2.22)$$

As a consequence of (2.20), we deduce that  $m_0(t)$  decays with time to a limit  $L_0 \geq 0$ .

*Third step. Limits for the  $m_k(t)$ .* We further write

$$\frac{d}{dt} \frac{m_{k+1}(t)}{m_k(t)} = \alpha \frac{m_0(t)}{m_k^2(t)} (m_{k+2}(t)m_k(t) - m_{k+1}^2(t)) \geq 0.$$

Therefore the above ratio has a limit and thus  $m_1(t) \rightarrow L_1$  as  $t \rightarrow \infty$ .

If  $L_1 = \infty$ , obviously from (2.20) we deduce  $L_0 = 0$  and the theorem is proved. We now assume, by opposition, that  $L_1 < +\infty$ ,  $L_0 > 0$  and thus  $m_2(t)$  has a limit  $L_2$ .

*Fourth step. A contradiction.* From the inequality (2.21), we deduce that  $L_1 = \frac{r}{1-\alpha}$  and this is a contradiction with the exact solution  $n(t, x)$  given in step (i).  $\square$

*Exercise.* In the case  $0 \leq x \leq A$ , prove that  $L_0 \neq 0$  and that  $n \rightarrow \rho\delta(x = A)$ . To do that check that in the proof above

$$L_2 L_0 = L_1^2.$$

*Exercise.* Prove that, in Theorem 2.3, the assumption  $\int_0^\infty x n^0(x) dx \geq \frac{r}{1-\alpha}$  is not needed because it will be true at some time  $t_0$ .

1. Assume  $m_1(t) < \frac{r}{1-\alpha}$  for all times, then prove that all  $m_k$  are increasing, and that  $m_1 \rightarrow \frac{r}{1-\alpha}$ , as  $t \rightarrow \infty$ .
2. Using the equation on  $m_k$  prove they are bounded, and have a limit as  $t \rightarrow \infty$ .
3. Prove a contradiction with the formula for the solution  $n(t, x)$  given in step (i).

## 2.4 Mutations

Until now we have seen ESS as the best adapted trait; in a population where initially all the traits are represented the ESS is the only one selected naturally.

We may ask what happens if a population initially does not represent all the traits but mutations may occur and create new mutants, that might be better adapted and thus be selected along with the selection principle we have already described, which themselves can mutate... etc The end of this chapter gives a formalism to address this question in rigorous mathematical terms.

In order to present this matter we need a more general framework than in Section 2.2, including a more general fitness in terms of the environment created by the total population. As we saw in previous sections, we can retain a structured population model that contains much of the previous examples while keeping some simplicity, writing

$$\begin{cases} \frac{d}{dt}n(t, x) = [b(x)Q_b(\varrho_b(t)) - d(x)Q_d(\varrho_d(t))]n(t, x), & x \in \mathbb{R}, t \geq 0, \\ \varrho_b(t) = \int_{\mathbb{R}} \psi_b(x)n(t, x)dx, & \varrho_d(t) = \int_{\mathbb{R}} \psi_d(x)n(t, x)dx, \\ n(t=0) = n^0(x) \geq 0. \end{cases} \quad (2.23)$$

Here the notation is the same as before and we assume that  $b, d, \psi_b, \psi_d \in C(\mathbb{R})$ ,  $Q_b, Q_d \in C^1(\mathbb{R}^+)$  and there are real numbers such that

$$Q'_b(\cdot) \leq 0, \quad Q'_d(\cdot) \geq 0, \quad (2.24)$$

$$\begin{cases} b_M \geq b(\cdot) \geq b_m > 0, & d_M \geq d(\cdot) \geq d_m > 0, \\ \psi_M \geq \psi_b(\cdot), \psi_d \geq \psi_m > 0, \end{cases} \quad (2.25)$$

$$\exists 0 < \varrho_m < \varrho_M \quad \text{such that} \quad \begin{cases} \max_{x \in \mathbb{R}} [b(x)Q_b(\psi_m \varrho_M) - d(x)Q_d(\psi_m \varrho_M)] < 0, \\ \min_{x \in \mathbb{R}} [b(x)Q_b(\psi_M \varrho_m) - d(x)Q_d(\psi_M \varrho_m)] > 0. \end{cases} \quad (2.26)$$

If reproduction is not completely faithful, an individual with trait  $y$  may generate offspring with trait  $x$ . Let  $K(x, y)$  be the corresponding mutation kernel. To simplify, we only consider the case  $K = K(x - y)$  with  $K(\cdot)$  a probability density. Also the whole birth process should not necessarily give mutations but we postulate so in order to avoid additional parameters which would not change the heart of the matter.

To incorporate these mutations in (2.23), we postulate the dynamics given by

$$\begin{cases} \frac{\partial}{\partial t}n(t, x) = Q_b(\varrho_b(t)) \int_{\mathbb{R}} b(y)K(x - y) n(t, y) dy - d(x)Q_d(\varrho_d(t))n(t, x), \\ \varrho_b(t) = \int_{\mathbb{R}} \psi_b(x)n(t, x)dx, & \varrho_d(t) = \int_{\mathbb{R}} \psi_d(x)n(t, x)dx, \\ n(t, 0) = n^0(x) \geq 0. \end{cases} \quad (2.27)$$

Our first claim is that this system is not much harder than an ordinary differential equation in terms of existence theory. We have

**Theorem 2.4.** *We assume (2.24)–(2.26), that  $K(\cdot) \in L^1(\mathbb{R})$  is a probability density and  $n^0 \in L^1$  satisfies  $\varrho^m \leq \varrho^0 \leq \varrho^M$ . Then the system (2.27) has a unique nonnegative solution such that  $n, \frac{\partial}{\partial t}n \in C(\mathbb{R}^+; L^1(\mathbb{R}))$  and for all  $t \geq 0$*

$$\varrho_m^0 \leq \varrho(t) \leq \varrho_M^0. \quad (2.28)$$

*Proof.* The method of proof is similar to that of the Cauchy–Lipschitz theorem and the main point is a global uniform bound. We begin with this bound and turn to the existence afterwards.

(i) A priori estimate. We first derive the inequalities (2.28) for nonnegative solutions. To do that, we integrate with respect to  $x$  the equation on  $n$  in (2.27) and notice that

$$\int_{\mathbb{R} \times \mathbb{R}} b(y)K(x-y) n(t, y) dy dx = \int_{\mathbb{R}} b(y) n(t, y) dy.$$

Therefore, we obtain

$$\begin{aligned} \frac{d}{dt}\varrho(t) &\leq \int_{y \in \mathbb{R}} [b(y)Q_b(\psi_m \varrho(t)) - d(y)Q_d(\psi_m \varrho(t))] n(t, y) dy \\ &\leq \varrho(t) \max_{y \in \mathbb{R}} [b(y)Q_b(\psi_m \varrho(t)) - d(y)Q_d(\psi_m \varrho(t))]. \end{aligned}$$

The inequality from above in (2.28) follows directly since, whenever  $\varrho(t)$  approaches  $\varrho_M$  (from below), then  $\max_{y \in \mathbb{R}} [b(y)Q_b(\psi_m \varrho_M) - d(y)Q_d(\psi_m \varrho_M)]$  becomes negative and then  $\varrho(t)$  decreases. A similar argument applies to prove the lower bound on  $\varrho(t)$ .

(ii) Existence. We prove existence using the Banach-Picard fixed point theorem. To do so, we consider the Banach space

$$X = C([0, T]; L^1(\mathbb{R})), \quad \|m\|_X := \sup_{0 \leq t \leq T} \|m(t)\|_{L^1(\mathbb{R})},$$

for some  $T > 0$  chosen later on. Then we define the closed subset

$$\mathcal{C} = \{m \in X, m \geq 0, \|m\|_X \leq C_\Phi\},$$

where  $C_\Phi$  is defined, for  $T$  small enough, so that

$$\varrho_M + Tb_M Q_b(0)C_\Phi \leq C_\Phi,$$

and we can always impose for instance  $C_\Phi \leq 2\varrho_M$ .

For  $m \in \mathcal{C}$ , we define  $R_b(t) = \int_{\mathbb{R}} \psi_b(x) m(t, x) dx$ ,  $R_d(t) = \int_{\mathbb{R}} \psi_d(x) m(t, x) dx$  and the solution ( $x$  by  $x$ ) to

$$\begin{cases} \frac{\partial}{\partial t} n(t, x) &= Q_b(R_b(t)) \int_{\mathbb{R}} b(y) K(x - y) m(t, y) dy - d(x) Q_d(R_d(t)) n(t, x), \\ n(t, 0) &= n^0(x) \geq 0. \end{cases} \quad (2.29)$$

Then, we claim that the operator

$$m \mapsto \Phi(m) := n,$$

has a unique fixed point in  $\mathcal{C}$ . We prove successively that

- (a)  $\Phi : \mathcal{C} \rightarrow \mathcal{C}$ ,
- (b)  $\Phi$  is a strong contraction for  $T$  small enough (depending only upon  $C_{\Phi}$ ).

Therefore we can apply the Banach–Picard theorem and deduce that  $\Phi$  has a unique fixed point in  $\mathcal{C}$  and the result is proved after iterating this argument on time-steps  $[T, 2T]$ ,  $[2T, 3T]$ ,  $\dots$ . Notice also that, because this fixed point  $n$  belongs to  $X$ ,  $\frac{\partial}{\partial t} n$  also belongs to  $X$ , and in each time-step the solution satisfies, thanks to step (i),  $\varrho_m \leq \varrho(t) \leq \varrho_M$ .

The point (a) follows for example from the solution formula

$$\begin{aligned} n(t, x) &= n^0(x) e^{-d(x) \int_0^t Q_d(R_d)} \\ &\quad + Q_b(R_b) \int_{s=0}^t \int_{\mathbb{R}} b(y) K(x - y) m(s, y) dy e^{-d(x) \int_s^t Q_d(R_d)} ds, \end{aligned}$$

which implies the first two properties  $n \geq 0$  and  $n \in C([0, T]; L^1(\mathbb{R}))$ . Then, we also have

$$\frac{d}{dt} n(t, x) \leq Q_b(R_b(t)) \int_{\mathbb{R}} b(y) K(x - y) m(s, y) dy.$$

Therefore we obtain for  $t \leq T$ ,

$$\begin{aligned} \int_{\mathbb{R}} n(t, x) dx &\leq \int_{\mathbb{R}} n^0(x) dx + \int_0^t Q_b(R_b(s)) \int_{\mathbb{R}} b(y) m(s, y) dy ds \\ &\leq \varrho_M + T b_M Q_b(0) \sup_{0 \leq s \leq T} R(s) \\ &= \varrho_M + T b_M Q_b(0) C_{\Phi} \\ &= C_{\Phi}. \end{aligned}$$

This proves indeed that  $\Phi(m)$  belongs to  $\mathcal{C}$ .

The point (b) is very standard. By subtraction of solutions for two different  $m$ , say  $m^1$  and  $m^2$ , we obtain

$$\begin{aligned} \frac{\partial}{\partial t}(n^1 - n^2) &= [Q_b(R_b^1) - Q_b(R_b^2)] \int b(y)K(x-y)m^1(y)dy \\ &\quad + Q_b(R_b^2) \int b(y)K(x-y)(m^1(y) - m^2(y))dy - d(x)[Q_d(R_d^1) - Q_d(R_d^2)]n^1 \\ &\quad - d(x)Q_d(R_d^2)(n^1 - n^2). \end{aligned}$$

From the a priori estimates of the step (a), the right-hand side is controlled by

$$C\|m^2 - m^1\|_{L^1(\mathbb{R})},$$

for some constant that uses  $C_\Phi$ . Therefore, since the initial data for  $n^2 - n^1$  vanishes,

$$\|m^2 - m^1\|_{L^1(\mathbb{R})} \leq CT\|m^2 - m^1\|_{L^1(\mathbb{R})},$$

and we have obtained the strong contraction property for  $T$  small enough. Notice that as mentioned before, this time  $T$  is independent of  $n^0$ .  $\square$

## 2.5 Small mutations: the constrained Hamilton–Jacobi equation

Along with the principles of adaptive dynamics, we now assume that mutations are small and we proceed to rescale accordingly the mutation model so as to arrive at a situation where a natural limit arises.

Hence, we introduce a (small) parameter  $\varepsilon > 0$  in the mutation kernel and we set (and assume)

$$K_\varepsilon(z) = \frac{1}{\varepsilon}K\left(\frac{z}{\varepsilon}\right), \quad K(\cdot) \geq 0, \quad \int K = 1, \quad \int e^{z^2}K(z)dz < \infty. \quad (2.30)$$

It is very clear that the dynamics described by the system (2.27) with  $K_\varepsilon$  in place of  $K$  is not interesting because the effect of mutations is too small. Indeed, we can write after the change of variable  $y = x - \varepsilon z$ ,

$$\begin{aligned} \int_{\mathbb{R}} b(y)K_\varepsilon(x-y) n(t, y) dy &= \int_{\mathbb{R}} b(x - \varepsilon z)K(z) n(t, x - \varepsilon z) dz \\ &\xrightarrow{\varepsilon \rightarrow 0} \int_{\mathbb{R}} b(x)K(z) n(t, x) dz = b(x) n(t, x). \end{aligned}$$

Therefore we end up with the original model (2.23).

To circumvent this, we observe the phenomena over a long time period, rescaling it with the same parameter  $\varepsilon$ . Hence we now consider the system

$$\begin{cases} \varepsilon \frac{\partial}{\partial t} n_\varepsilon(t, x) = Q_b(\varrho_{b,\varepsilon}(t)) \int_{\mathbb{R}} b(y) K_\varepsilon(x - y) n_\varepsilon(t, y) dy - Q_d(\varrho_{d,\varepsilon}(t)) n_\varepsilon(t, x), \\ \varrho_{b,\varepsilon}(t) = \int_{\mathbb{R}} \psi_b(x) n_\varepsilon(t, x) dx, \quad \varrho_{d,\varepsilon}(t) = \int_{\mathbb{R}} \psi_d(x) n_\varepsilon(t, x) dx, \\ n_\varepsilon(t = 0) = n_\varepsilon^0(x) \geq 0. \end{cases} \quad (2.31)$$

We explain below that, on this new time scale, a (or several in special cases) dominant trait  $\bar{x}(t)$  is selected (following the selection principle of Sections 2.1 and 2.2). But now, this dominant trait evolves according to an interesting, and not explicit at all, dynamics. This means that, typically (but not always), we have

$$n_\varepsilon(t, x) \xrightarrow{\varepsilon \rightarrow 0} \bar{\varrho}(t) \delta(x - \bar{x}(t)). \quad (2.32)$$

Such a limiting population is called *monomorphic*. A polymorphic population corresponds to the sum of several Dirac masses in this limit. This arises for instance in a chemostat with several nutrients ([83]). We will refer to this (these) concentration point(s) as the dominant trait(s).

In order to derive the limiting dynamics, we assume a monomorphic initial population

$$n_\varepsilon(t, x) = e^{\varphi_\varepsilon(t, x)/\varepsilon}, \quad n_\varepsilon^0(x) = e^{\varphi_\varepsilon^0(x)/\varepsilon}, \quad (2.33)$$

such that, as  $\varepsilon$  vanishes, there holds

$$\begin{cases} \int_{\mathbb{R}} n_\varepsilon^0(x) dx \rightarrow M^0 > 0, \quad \varphi_\varepsilon^0 \rightarrow \varphi^0 \leq 0 \quad \text{uniformly in } \mathbb{R}, \\ \max_{x \in \mathbb{R}} \varphi^0(x) = 0 = \varphi(\bar{x}^0) \quad \text{for a single } \bar{x}^0. \end{cases} \quad (2.34)$$

As an example, we might have in mind a gaussian distribution of traits in the population

$$n_\varepsilon^0(x) = \frac{1}{\sqrt{2\pi\varepsilon}} e^{-|x|^2/2\varepsilon}, \quad \varphi_\varepsilon^0(x) = -|x|^2 - \varepsilon \ln(2\pi\varepsilon).$$

It clearly shows that the uniqueness of the maximum point of  $\varphi^0$  is equivalent to the monomorphic Dirac concentration.

The reason why the change of unknown (2.33) is useful can be guessed from writing the equation on  $\varphi_\varepsilon$ :

$$\begin{aligned} \frac{\partial \varphi_\varepsilon(t, x)}{\partial t} &= e^{-\frac{\varphi_\varepsilon(t, x)}{\varepsilon}} Q_b(\varrho_{b,\varepsilon}(t)) \int_{\mathbb{R}} b(y) K_\varepsilon(x - y) e^{\frac{\varphi_\varepsilon(t, y)}{\varepsilon}} dy - Q_d(\varrho_{d,\varepsilon}(t)) d(x) \\ &= Q_b(\varrho_{b,\varepsilon}(t)) \int_{\mathbb{R}} K(z) b(x + \varepsilon z) e^{\frac{\varphi_\varepsilon(t, x + \varepsilon z) - \varphi_\varepsilon(t, x)}{\varepsilon}} dz - Q_d(\varrho_{d,\varepsilon}(t)) d(x) \\ &\xrightarrow{\varepsilon \rightarrow 0} Q_b(\varrho_b(t)) b(x) \int_{\mathbb{R}} K(z) e^{z \frac{\partial}{\partial x} \varphi(t, x)} dz - Q_d(\varrho_d(t)) d(x). \end{aligned}$$

Therefore the limiting function  $\varphi$  should satisfy the *constrained Hamilton–Jacobi equation*

$$\begin{cases} \frac{\partial}{\partial t}\varphi(t, x) = Q_b(\varrho_b(t))b(x)H\left(\frac{\partial}{\partial x}\varphi(t, x)\right) - Q_d(\varrho_d(t))d(x), & t \geq 0, x \in \mathbb{R}, \\ \max_{x \in \mathbb{R}} \varphi(t, x) = 0, & \forall t \geq 0, \\ \varphi(t = 0, x) = \varphi^0(x) \leq 0 \end{cases} \quad (2.35)$$

with

$$H(p) := \int_{\mathbb{R}} K(z) e^{zp} dz. \quad (2.36)$$

This function  $H(\cdot)$  is called the *hamiltonian* and has the properties

$$\begin{cases} H(p) > 0, & H(0) = \int_{\mathbb{R}} K(z) dz = 1, & H''(p) = \int_{\mathbb{R}} K(z) z^2 e^{zp} dz > 0, \\ H'(0) = \int_{\mathbb{R}} K(z) z dz = 0 & \text{for } K(\cdot) \text{ even,} \end{cases} \quad (2.37)$$

in other words,  $H(\cdot)$  is a positive convex and even hamiltonian for  $K$  even. Finally the constraint  $\max_{x \in \mathbb{R}} \varphi(t, x) = 0$  is equivalent to saying that  $n_e$  has bounded mass and non-extinction.

The problem (2.35)–(2.36) can be understood as follows. The unknowns are  $\varphi(t, x)$  and the density  $\bar{\varrho}(t)$  in (2.32). But they do not play the same role. The parameter  $\bar{\varrho}(t)$  defines the *Lagrange multipliers*  $Q_b(\bar{\varrho}_b(t))$  and  $Q_d(\bar{\varrho}_d(t))$  for the constraint that  $\max \varphi(t, \cdot) = 0$ .

When the maximum of  $\varphi(t, \cdot)$  is unique as initially (this is a monomorphic situation), we can recover the dominant trait in (2.32) by

$$\max_{x \in \mathbb{R}} \varphi(t, x) = 0 = \varphi(t, \bar{x}(t)).$$

Also the functions  $Q_b(\bar{\varrho}_b(t))$  and  $Q_d(\bar{\varrho}_d(t))$  are then given by

$$\bar{\varrho}_b(t) = \psi_b(\bar{x}(t))\bar{\varrho}(t), \quad \bar{\varrho}_d(t) = \psi_d(\bar{x}(t))\bar{\varrho}(t).$$

More details are given at the end of Section 2.7 on the way to use these formulas.

The asymptotic, existence and uniqueness theory for deriving rigorously the constrained Hamilton–Jacobi equation (2.36) has been developed in [18] in the case  $\psi_b = \psi_d$ . In the general case uniqueness is an open question. The main difficulty is the interpretation of the two Lagrange multipliers  $Q_b(\bar{\varrho}_b(t))$  and  $Q_d(\bar{\varrho}_d(t))$  that makes it possible to contain the dimorphic situation of [83] where an additional uniqueness criteria is needed, and this is an open problem to find it. Again, we refer to Section 2.7 for more details.

We conclude this section by a remark: this asymptotic formalism is robust and can be extended to many situations under investigation. For instance it can be extended to systems as in [51] where a population with juveniles and adults is carried out, leading to a much more complicated hamiltonian.

## 2.6 A simple case of constrained Hamilton–Jacobi equation

A particularly simple situation is that of Section 2.1. Then  $Q_b \equiv 1$ ,  $d \equiv 1$   $\psi_d \equiv 1$  and  $Q_d(u) = u$ . Then the constrained Hamilton–Jacobi becomes

$$\begin{cases} \frac{\partial}{\partial t} \varphi(t, x) = b(x)H\left(\frac{\partial}{\partial x} \varphi(t, x)\right) - \bar{\varrho}(t), & t \geq 0, x \in \mathbb{R}, \\ \max_{x \in \mathbb{R}} \varphi(t, x) = 0 = \varphi(t, \bar{x}(t)), & \forall t \geq 0, \\ \varphi(t = 0, x) = \varphi^0(x) \leq 0. \end{cases} \quad (2.38)$$

Reflecting the simplicity of the model, this situation is particularly simple because one can reduce it to a more classical Hamilton–Jacobi equation. To do this, we introduce two new unknown functions

$$R(t) = \int_0^t \varrho(s) ds, \quad \psi(t, x) = \varphi(t, x) + R(t).$$

Then, we arrive at the Hamilton–Jacobi equation (without a constraint) on  $\psi$ ,

$$\begin{cases} \frac{\partial}{\partial t} \psi(t, x) = b(x)H\left(\frac{\partial}{\partial x} \psi(t, x)\right), & t \geq 0, x \in \mathbb{R}, \\ \psi(t = 0, x) = \varphi^0(x). \end{cases} \quad (2.39)$$

This is the reason why one can derive a rigorous result that we state now but that we prove in Section 2.8. This is not possible for more complex dynamics as in [83], see also Section 2.9.

We use the notation

$$R_\varepsilon(t) = \int_0^t \varrho_\varepsilon(s) ds, \quad \psi_\varepsilon(t, x) = \varphi_\varepsilon(t, x) + R_\varepsilon(t). \quad (2.40)$$

**Proposition 2.1.** *Assume that (2.30) holds. We have, for all  $t \geq 0$ , the bounds*

$$\min \left( \min_{y \in \mathbb{R}} b(y), \varrho_\varepsilon^0 \right) \leq \varrho_\varepsilon(t) \leq \max \left( \max_{y \in \mathbb{R}} b(y), \varrho_\varepsilon^0 \right). \quad (2.41)$$

If the initial data satisfies  $\varphi_\varepsilon^0(x) \leq -|x| + C_\varepsilon^0$ , then,

$$\psi_\varepsilon(t, x) \leq -|x| + C_\varepsilon^0 + t \max b \max_{|p| \leq 1} H(p), \quad (2.42)$$

$$\left| \frac{\partial}{\partial t} \psi_\varepsilon(t, x) \right| \leq \max_{y \in \mathbb{R}} |b(y)| \max_{y \in \mathbb{R}} H\left(\frac{\partial}{\partial y} \varphi_\varepsilon^0(y)\right). \quad (2.43)$$

**Theorem 2.5.** *We assume that (2.30), (2.34) hold and uniform bounds on the right hand sides in Proposition 2.1, on  $\frac{\partial}{\partial x} \psi_\varepsilon^0(y)$  and on  $\frac{\partial^2 b}{\partial x^2}$ . Then, after extracting a subsequence, we have*

$$\psi_\varepsilon(t, x) \xrightarrow{\varepsilon \rightarrow 0} \psi(t, x), \quad \varphi_\varepsilon(t, x) \xrightarrow{\varepsilon \rightarrow 0} \varphi(t, x) \quad \text{locally uniformly,}$$

$\psi$  satisfies the Hamilton–Jacobi equation (2.39) in the sense of a viscosity solution and for all  $t \geq 0$ ,  $\max_{x \in \mathbb{R}} \varphi(t, x) = 0$ . Also the inequalities in Proposition 2.1 hold for  $\psi(t, x)$ .

The notion of viscosity solutions for Hamilton–Jacobi equations has been introduced as a uniqueness criteria by Crandall and Lions (see [69] and the references therein). Its definition, motivation and main properties can be found in several recent surveys and books, see for instance [16, 69, 14, 98].

Theorem 2.5 gives a meaning to the system (2.35), simply the equation on  $\varphi$  has to be understood as an equation on  $\psi$ . Again, we refer to [18] for uniqueness.

## 2.7 A partial canonical equation and dynamics of the dominant trait

One can try to derive, from the Hamilton–Jacobi equation (2.35)–(2.36), an equation for the evolution of the dominant trait  $\bar{x}(t)$ . Such an equation, called *canonical equation* was proposed by Dieckmann and Law [72] for a model based on a stochastic process. We derive such an equation in our context and then apply it to monomorphic and dimorphic examples.

### 2.7.1 Derivation of a canonical equation

As we explain now, we can do it in our formalism but we can only derive partial information, namely the sense of variation of the trait. To do so, we assume here that

$$\frac{\partial H(0)}{\partial p} = 0, \quad \text{i.e., } K(\cdot) \text{ is even.} \quad (2.44)$$

Our purpose is to prove

**Lemma 2.1.** *Assuming (2.44), in smoothness regimes, a dominant trait  $\bar{x}(t)$  evolves according to the dynamics (canonical equation)*

$$Q_b(\bar{\varrho}_b(t))b(\bar{x}(t)) = Q_d(\bar{\varrho}_d(t))d(\bar{x}(t)), \quad (2.45)$$

$$\frac{d}{dt}\bar{x}(t) = \left( -\frac{\partial^2 \varphi(t, \bar{x}(t))}{\partial x^2} \right)^{-1} \left[ Q_b(\bar{\varrho}_b(t))\frac{\partial b(\bar{x}(t))}{\partial x} - Q_d(\bar{\varrho}_d(t))\frac{\partial d(\bar{x}(t))}{\partial x} \right], \quad (2.46)$$

and because  $\varphi$  achieves a maximum at  $\bar{x}(t)$ , we have  $\frac{\partial^2}{\partial x^2}\varphi(t, \bar{x}(t)) \leq 0$ .

In the terminology of adaptive dynamics, the quantity in the bracket is called the *selection gradient*.

*Proof.* Departing from the property that a dominant trait satisfies  $\varphi(t, \bar{x}(t)) = 0$ , and  $\varphi(t, x) \leq 0$ , we also deduce that

$$\frac{\partial}{\partial t}\varphi(t, \bar{x}(t)) = 0, \quad \frac{\partial}{\partial x}\varphi(t, \bar{x}(t)) = 0, \quad \frac{\partial^2}{\partial x^2}\varphi(t, \bar{x}(t)) \leq 0.$$

From the equation (2.35) we deduce directly (2.45).

We can also calculate

$$\begin{aligned} 0 &= \frac{d}{dt} \left[ \frac{\partial}{\partial x}\varphi(t, \bar{x}(t)) \right] \\ &= \frac{\partial}{\partial t} \frac{\partial}{\partial x}\varphi(t, \bar{x}(t)) + \dot{\bar{x}}(t) \frac{\partial^2}{\partial x^2}\varphi(t, \bar{x}(t)). \end{aligned}$$

Next, we differentiate in  $x$  the equation (2.35) and arrive at

$$\frac{\partial^2 \varphi(t, x)}{\partial t \partial x} = Q_b(\bar{\varrho}_b(t))\frac{\partial b(x)}{\partial x}H\left(\frac{\partial}{\partial x}\varphi\right) + b(x)H_p\left(\frac{\partial}{\partial x}\varphi\right)\frac{\partial^2 \varphi(t, x)}{\partial x^2} - Q_d(\bar{\varrho}_d(t))\frac{\partial d(x)}{\partial x}.$$

At the point  $\bar{x}(t)$  we have, along with (2.44),  $\frac{\partial H(0)}{\partial p} = 0$  and thus we obtain simply

$$\frac{\partial}{\partial t} \frac{\partial}{\partial x}\varphi(t, \bar{x}(t)) = Q_b(\bar{\varrho}_b(t))\frac{\partial b(\bar{x}(t))}{\partial x} - Q_d(\bar{\varrho}_d(t))\frac{\partial d(\bar{x}(t))}{\partial x}.$$

With the above information on  $\dot{\bar{x}}(t)$  we arrive at (2.46).  $\square$

Notice that one cannot close this by another equation on  $\frac{\partial^2 \varphi(t, \bar{x}(t))}{\partial x^2}$ . Indeed such an equation involves a higher order derivative  $\frac{\partial^3 \varphi(t, \bar{x}(t))}{\partial x^3}$  and so on. Also, it might turn out that  $\frac{\partial^2 \varphi(t, \bar{x}(t))}{\partial x^2}$  vanishes or that  $\varphi$  achieves several maxima at some times. These are singularities where the canonical equation does not apply.

Nevertheless we can infer from the equation (2.46) the dynamics of the dominant trait and we distinguish the monomorphic and dimorphic cases.

### 2.7.2 Monomorphic adaptive evolution

We assume that the population is monomorphic, i.e., there is a single trait  $\bar{x}(t)$ , at each time, such that  $\varphi(t, \bar{x}(t)) = 0$ , or in other words that the limit of  $n_\varepsilon$  is a single Dirac mass. Then, we obtain in (2.31) that

$$\bar{\varrho}_b(t) = \bar{\varrho}(t)\psi_b(\bar{x}(t)), \quad \bar{\varrho}_d(t) = \bar{\varrho}(t)\psi_d(\bar{x}(t)).$$

Therefore, using (2.45)

$$Q_b(\bar{\varrho}(t)\psi_b(\bar{x}(t)))b(\bar{x}(t)) = Q_d(\bar{\varrho}(t)\psi_d(\bar{x}(t)))d(\bar{x}(t)) \quad \forall t \geq 0.$$

This relation defines (say by continuity)  $\bar{\varrho}(t)$  as a function  $R(\bar{x}(t))$  which can be inserted in (2.46) to obtain a differential equation on  $\bar{x}(t)$  (once  $D^2\varphi$  is known).

At least, we can use the canonical equation (2.46) to deduce that  $\bar{x}(t)$  evolves in the direction of increasing values in  $x$  of the quantity

$$Q_b(\bar{\varrho}_b(t))b(x) - Q_d(\bar{\varrho}_d(t))d(x).$$

This explains the denomination *selection gradient* for its  $x$  derivative. Therefore,  $(\bar{\varrho}_b(t), \bar{\varrho}_d(t), \bar{x}(t))$  will generically reach a point  $(\bar{\varrho}_b^*, \bar{\varrho}_d^*, \bar{x}^*)$ , where

$$0 = Q_b(\bar{\varrho}_b^*)b(\bar{x}^*) - Q_d(\bar{\varrho}_d^*)d(\bar{x}^*) = \max_{x \in \mathbb{R}} [Q_b(\bar{\varrho}_b^*)b(x) - Q_d(\bar{\varrho}_d^*)d(x)], \quad (2.47)$$

with

$$\bar{\varrho}_b^* = \psi_b(\bar{x}^*)\bar{\varrho}^*, \quad \bar{\varrho}_d^* = \psi_d(\bar{x}^*)\bar{\varrho}^*. \quad (2.48)$$

These two formulas (2.47)–(2.48) on the two unknowns  $\bar{x}^*$  and  $\bar{\varrho}^*$  are typical characterizations of the ESS as we have already seen it rigorously in specific examples (see Sections 2.1 and 2.2).

We recall that the logic behind these formula has been explained in Section 2.2, when  $\psi_b$  and  $\psi_d$  are constant. Then the function

$$\rho \mapsto \max_{x \in \mathbb{R}} [Q_b(\psi_b \bar{\varrho})b(x) - Q_d(\psi_d \bar{\varrho})d(x)]$$

is monotonic (when  $Q_b$  is decreasing and  $Q_d$  is increasing) and can vanish at a single point  $\bar{\varrho}^*$ . Then, if  $b/d$  is monotonic, then the  $\max_x$  is attained itself at a single point  $x^*$ .

### 2.7.3 Dimorphic adaptive evolution

A dimorphic population can occur if there are two dominant traits

$$n_\varepsilon(t, x) \rightarrow \bar{\varrho}_1(t)\delta(x - \bar{x}_1(t)) + \bar{\varrho}_2(t)\delta(x - \bar{x}_2(t)),$$

which also means

$$0 = \max_x \varphi(t, x) = \varphi(t, \bar{x}_1(t)) = \varphi(t, \bar{x}_2(t)).$$

According to the relation (2.45) this means that we should have

$$\frac{Q_b(\bar{\varrho}_b(t))}{Q_d(\bar{\varrho}_d(t))} = \frac{d(\bar{x}_1(t))}{b(\bar{x}_1(t))} = \frac{d(\bar{x}_2(t))}{b(\bar{x}_2(t))} := R(t).$$

In other words, a condition for dimorphism is that  $b/d$  is not one-to-one. If it has a ‘parabolic profile’, the above ratio  $R(t)$  determines  $\bar{x}_1(t)$  and  $\bar{x}_2(t)$ . Which themselves constrain  $\varrho_1(t)$  and  $\varrho_2(t)$  by a first relation

$$Q_b(\bar{\varrho}_1(t))\psi_b(\bar{x}_1(t)) + \bar{\varrho}_2(t)\psi_b(\bar{x}_2(t)) = R(t)Q_b(\bar{\varrho}_1(t))\psi_b(\bar{x}_1(t)) + \bar{\varrho}_2(t)\psi_b(\bar{x}_2(t)).$$

This also means that the two differential equations given by (2.46) are in fact reduced to the same dynamics (say for instance on  $\bar{x}_1(t)$ ) through the inverse of the second derivative of  $\varphi$ .

The evolution of two such traits is therefore highly constrained and it is not as easy as before to characterize the dimorphic ESS. The same logic leads to

$$0 = \max_x [Q_b(\bar{\varrho}_b^*)b(x) - Q_d(\bar{\varrho}_d^*)d(x)] = Q_b(\bar{\varrho}_b^*)b(\bar{x}_i^*) - Q_d(\bar{\varrho}_d^*)d(\bar{x}_i^*), \quad i = 1, 2,$$

$$\bar{\varrho}_b^* = \bar{\varrho}_1^*\psi_b(x_1^*) + \bar{\varrho}_2^*\psi_b(x_2^*),$$

$$\bar{\varrho}_d^* = \bar{\varrho}_1^*\psi_d(x_1^*) + \bar{\varrho}_2^*\psi_d(x_2^*).$$

Again, we can analyze these formulae in the case where  $\psi_b$  and  $\psi_d$  are constant (and these constants should be different!). Then  $\bar{\varrho}_b^*$  and  $\bar{\varrho}_d^*$  are two independent Lagrange multipliers and we may expect a one-parameter family of dimorphic ESS.

Also, the constrained Hamilton–Jacobi equation (2.35) is clearly not enough to decide between the monomorphic interpretation and the dimorphic equation and this is a reason for non-uniqueness. In the monomorphic case  $\varphi(t, \cdot)$  achieves a single maximum with a single Lagrange multiplier in the interpretation  $n(t, x) = \varrho(t)\delta(x - \bar{x}(t))$ . In the dimorphic case  $\varphi(t, \cdot)$  achieves two maxima that require two independent Lagrange multipliers provided by the interpretation  $n(t, x) = \varrho_1(t)\delta(x - \bar{x}_1(t)) + \varrho_2(t)\delta(x - \bar{x}_2(t))$ . It is an open problem to find a criterion for uniqueness.

### 2.7.4 Adaptive evolution: cannibalism

As an example we can treat the simple model of cannibalism from Section 2.3. This corresponds, after rescaling and introducing mutations, to the equation

$$\begin{cases} \varepsilon \frac{\partial}{\partial t} n_\varepsilon(t, x) = \int_{\mathbb{R}} (r + \alpha y \varrho_\varepsilon(t)) K_\varepsilon(x - y) n_\varepsilon(t, y) dy - \varrho_{d,\varepsilon}(t) n_\varepsilon(t, x), \\ \varrho_\varepsilon(t) = \int_{\mathbb{R}} n_\varepsilon(t, x) dx, \quad \varrho_{d,\varepsilon}(t) = \int_{\mathbb{R}} x n_\varepsilon(t, x) dx, \\ n_\varepsilon(t = 0) = n_\varepsilon^0(x) \geq 0. \end{cases} \quad (2.49)$$

The limit  $\varepsilon \rightarrow 0$  gives the constrained Hamilton–Jacobi equation

$$\frac{\partial}{\partial t} \varphi(t, x) = (r + \alpha x \varrho(t)) H(\nabla \varphi) - \varrho_d(t). \quad (2.50)$$

Therefore the dominant trait should satisfy the relation (2.45) which here reads

$$r + \alpha \bar{x}(t) \varrho(t) = \varrho_d(t).$$

Since there is a single possible solution  $\bar{x}(t)$ , the population can only be monomorphic (see [79] for more elaborate models with possibly dimorphism). Also as we already had it intuitively from Section 2.3, the canonical equation (2.46) gives the evolution

$$\frac{d}{dt} \bar{x}(t) = \left( - \frac{\partial^2 \varphi(t, \bar{x}(t))}{\partial x^2} \right)^{-1} \alpha \varrho(t) > 0,$$

(here the *selection gradient* is simply  $\alpha \varrho(t)$ ) which tells us that the trait increases (to infinity in principle).

### 2.7.5 Adaptive evolution: chemostat with two nutrients

In the case of two nutrients with simplified predation function and for fast response of the nutrients, the system (2.1) becomes

$$\begin{cases} S_1(t) = \frac{S_{01}}{1 + \int \eta_1(x) n(t, x) dx}, \quad t \geq 0, x > 0, \\ S_2(t) = \frac{S_{02}}{1 + \int \eta_2(x) n(t, x) dx}, \\ \frac{\partial}{\partial t} n(t, x) = -R n(t, x) + \int (\eta_1(y) S_1(t) + \eta_2(y) S_2(t)) n(t, y) K_\varepsilon(x - y) dy, \\ n(t = 0, x) = n^0(x) > 0, n^0 \in L^1 \cap L^\infty(\mathbb{R}^+). \end{cases} \quad (2.51)$$

One can deduce the constrained Hamilton–Jacobi equation

$$\frac{\partial}{\partial t} \varphi(t, x) = -R + (\eta_1(x) S_1(t) + \eta_2(x) S_2(t)) H(\nabla \varphi).$$

In the present case the *selection gradient* is  $\eta'_1(x)S_1(t) + \eta'_2(x)S_2(t)$ .

Therefore the dominant traits should satisfy the relation (2.45) which here reads

$$R = \eta_1(x_1(t))S_1(t) + \eta_2(x_1(t))S_2(t) = \eta_1(x_2(t))S_1(t) + \eta_2(x_2(t))S_2(t).$$

It is possible to achieve dimorphism if the function  $\eta_1(x)S_1 + \eta_2(x)S_2$  has a ‘parabolic profile’. This is possible for instance when  $\eta_j$  are positive convex functions with  $\eta_1$  decreasing and  $\eta_2$  increasing.

## 2.8 A rigorous derivation of the Hamilton–Jacobi equation

This section is devoted to the proofs of the results stated in Section 2.5.

*Proof of Proposition 2.1.* The first statement, (2.41), is nothing but the a priori estimate (2.28). For the second statement, (2.42), we write the Hamilton–Jacobi equation on  $\psi_\varepsilon$

$$\begin{cases} \frac{\partial}{\partial t}\psi_\varepsilon(t, x) = \int b(x + \varepsilon z) e^{(\psi_\varepsilon(t, x + \varepsilon z) - \psi_\varepsilon(t, x))/\varepsilon} M(z) dz := K_\varepsilon(x, \psi_\varepsilon(t, \cdot)), \\ \psi_\varepsilon(t = 0, x) = \varphi_\varepsilon^0(x). \end{cases} \quad (2.52)$$

We just notice that  $\bar{\psi}_\varepsilon(t) := -|x| + C_\varepsilon^0 + t \max b \max_{|p| \leq 1} H(p)$  is a supersolution to this Hamilton–Jacobi equation (and also to (2.39)) since

$$\frac{\partial}{\partial t}\bar{\psi}_\varepsilon(t, x) = \max b \max_{|p| \leq 1} H(p) \geq K_\varepsilon(x, \bar{\psi}_\varepsilon(t, \cdot)) \quad \text{and} \quad \geq b(x)H\left(\frac{\partial}{\partial x}\bar{\psi}_\varepsilon(t, x)\right).$$

For the last statement, (2.43), we consider  $\Psi_\varepsilon(t, x) = \frac{\partial}{\partial t}\psi_\varepsilon(t, x)$  and differentiate the equation (2.52) to obtain

$$\frac{\partial}{\partial t}\Psi_\varepsilon(t, x) = \frac{1}{\varepsilon} \int b(x + \varepsilon z) e^{(\psi_\varepsilon(t, x + \varepsilon z) - \psi_\varepsilon(t, x))/\varepsilon} (\Psi_\varepsilon(t, x + \varepsilon z) - \Psi_\varepsilon(t, x)) M(z) dz.$$

At the maximum point,  $\Psi_\varepsilon(t, x_0) = \max_{x \in \mathbb{R}} \Psi_\varepsilon(t, x)$ , we deduce that

$$\frac{\partial}{\partial t}\Psi_\varepsilon(t, x_0) \leq 0,$$

and thus  $\max_{x \in \mathbb{R}} \Psi_\varepsilon(t, x) \leq \max_{x \in \mathbb{R}} \Psi_\varepsilon(t = 0, x)$ . The same argument leads to control the *minimum*. Finally, we use the equation (2.52) at  $t = 0$  to obtain

$$\begin{aligned} |\Psi_\varepsilon(t = 0, x)| &= \int b(x + \varepsilon z) e^{(\varphi_\varepsilon^0(x + \varepsilon z) - \varphi_\varepsilon^0(x))/\varepsilon} M(z) dz \\ &\leq \max_{y \in \mathbb{R}} |b(y)| \max_{y \in \mathbb{R}} H\left(\frac{\partial}{\partial x}\varphi_\varepsilon^0(y)\right), \end{aligned}$$

and the third statement is proved.  $\square$

*Proof of Theorem 2.5.* As a first step, we now derive a bound on the  $x$  derivative of  $\psi_\varepsilon$ . To do so we fix a time  $T > 0$  and set  $\Phi_\varepsilon(t, x) = \frac{\partial}{\partial x} \psi_\varepsilon(t, x)$ . We write the equation (2.52) as

$$\frac{\partial}{\partial t} \frac{\psi_\varepsilon(t, x)}{b(x)} = \int \frac{b(x + \varepsilon z)}{b(x)} e^{(\psi_\varepsilon(t, x + \varepsilon z) - \psi_\varepsilon(t, x))/\varepsilon} M(z) dz.$$

Next, we compute

$$\begin{aligned} \frac{\partial}{\partial t} \frac{\Phi_\varepsilon(t, x)}{b(x)} &= \frac{\partial}{\partial t} \psi_\varepsilon(t, x) \frac{\partial b(x)}{\partial x} / b(x)^2 \\ &+ \int \frac{\partial}{\partial x} \left( \frac{b(x + \varepsilon z)}{b(x)} \right) e^{(\psi_\varepsilon(t, x + \varepsilon z) - \psi_\varepsilon(t, x))/\varepsilon} M(z) dz \\ &+ \frac{1}{\varepsilon} \int \frac{b(x + \varepsilon z)}{b(x)} e^{(\psi_\varepsilon(t, x + \varepsilon z) - \psi_\varepsilon(t, x))/\varepsilon} (\Phi_\varepsilon(t, x + \varepsilon z) - \Phi_\varepsilon(t, x)) M(z) dz. \end{aligned}$$

Again, we consider the maximum point

$$\Phi_\varepsilon(t, x_0) = \max_{y \in \mathbb{R}} |\Phi_\varepsilon(t, y)| := Q_\varepsilon(t),$$

and estimate

$$\left| \frac{\partial}{\partial x} \left( \frac{b(x + \varepsilon z)}{b(x)} \right) \right| \leq \varepsilon \left[ \max_{y \in \mathbb{R}} \left| \frac{\partial^2 b(y)}{\partial y^2} \right| \max_{y \in \mathbb{R}} |b(y)| + \max_{y \in \mathbb{R}} \left| \frac{\partial b(y)}{\partial y} \right|^2 \right] / b(x)^2.$$

Multiplying by  $b(x)$  we deduce again, because at the maximum point the last term in the right hand-side is non-positive, and at the minimum point it is nonnegative, that

$$\frac{d}{dt} Q_\varepsilon(t) \leq \max_{y \in \mathbb{R}} \left[ \left| \frac{\partial}{\partial t} \varphi_\varepsilon^0(y) \right| \left| \frac{\partial b(y)}{\partial x} / b(y) \right| \right] + \varepsilon Q_b e^{Q_\varepsilon(t)} \int M(z) dz$$

with  $Q_b = [\max_{y \in \mathbb{R}} |\frac{\partial^2 b(y)}{\partial y^2}| \max_{y \in \mathbb{R}} |b(y)| + \max_{y \in \mathbb{R}} |\frac{\partial b(y)}{\partial y}|^2] / [\min_{y \in \mathbb{R}} b(y)]$ . This provides a uniform bound on  $Q_\varepsilon(t)$  which proves that  $\psi_\varepsilon$  is locally compact thanks to the Ascoli–Arzela Theorem.

Therefore we may extract subsequences such that  $R_\varepsilon(t)$  (which is uniformly lipschitz continuous) and  $\psi_\varepsilon(t, x)$  converge locally uniformly. This is enough to pass to the limit in the viscosity sense in the equation (2.52) and to recover (2.39) (see [16, 69, 14, 98] for basic handling of the Hamilton–Jacobi equation). Finally, the statement  $\max_{x \in \mathbb{R}} \varphi(t, x) = 0$  follows from the bound (2.41) which would not hold if  $\max_{x \in \mathbb{R}} \varphi(t, x)$  would be positive or negative in view of the definition  $n_\varepsilon = e^{\varphi_\varepsilon/\varepsilon}$ .  $\square$

## 2.9 Numerical results and examples of adaptive evolution

In this section we present several numerical experiments that illustrate possible evolutions of the dominant traits and that allow comparison between the direct simulation of the structured population equations (2.31) and the constrained Hamilton–Jacobi equation (2.35). The advantage of the latter is that it requires very few points compared to the former which require several mesh points per  $\varepsilon$ .

We first present several possible behaviors for monomorphic situations. They correspond to the general formalism (2.35) with  $Q_d \equiv 1$  and  $Q_b = \varrho = \int n(t, x)dx$  with a trait which is chosen, for numerical treatment in  $[0, 1]$ , and we present various cases of function  $b(x)$ . A continuous evolution is presented in Figure 2.1 for the function  $b(x) = .5 + x(2 - x)$ .

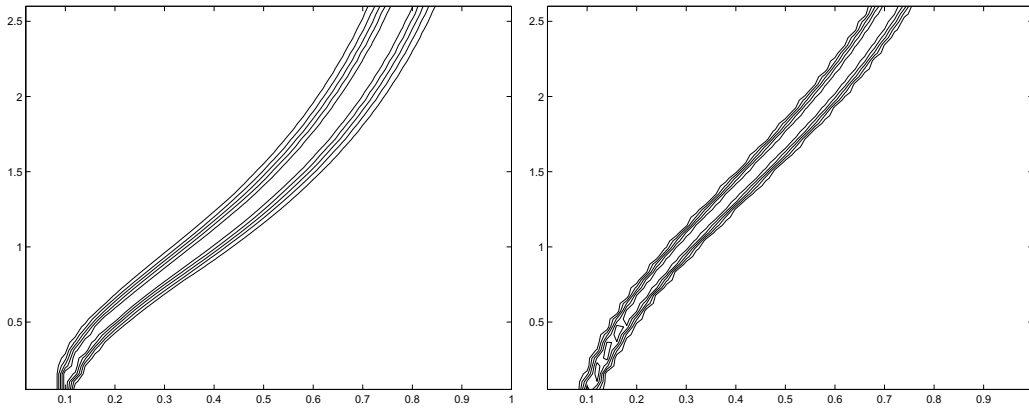


Figure 2.1: CONTINUOUS EVOLUTION OF A TRAIT ACCORDING TO DIRECT SIMULATION OF THE STRUCTURED POPULATION EQUATIONS (2.31) (LEFT) AND THE CONSTRAINED HAMILTON–JACOBI EQUATION (2.35) (RIGHT). THE HORIZONTAL COORDINATE IS THE TRAIT  $x$  AND THE VERTICAL COORDINATE IS TIME.

In Figure 2.2 we illustrate the possibility of a discontinuity of the dominant trait and this corresponds to the choice  $b(x) = \min(.45 + x^2, .55 + .4x)$ , with the same values of  $Q_b$  and  $Q_d$  as before. The main interest of this example is to show that a mere differential equation cannot be enough to describe the evolution of the dominant trait.

Finally in Figure 2.3, we show the bifurcation from a monomorphic to a dimorphic population in the case of the chemostat with two nutrients (2.51), following the theory and example developed in [83].

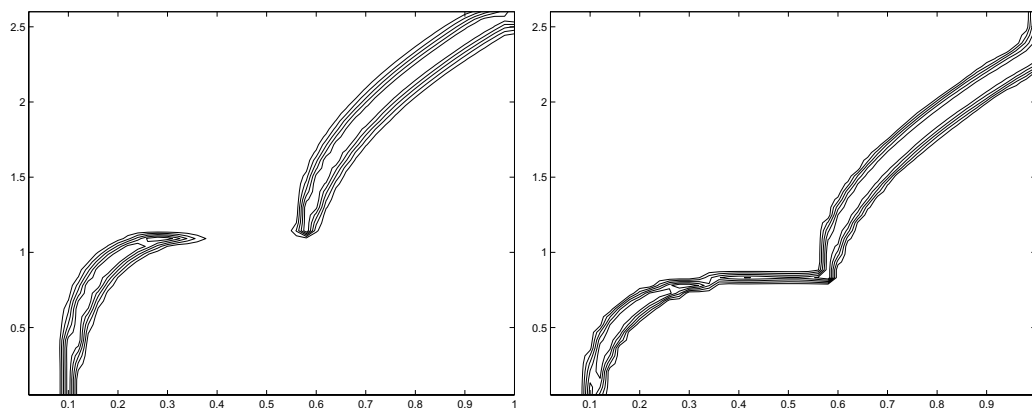


Figure 2.2: SAME AS FIGURE 2.1 FOR A TRAIT EVOLUTION WITH A JUMP IN A MONOMORPHIC POPULATION (THIS SHOWS THAT A DIFFERENTIAL EQUATION CANNOT BE ENOUGH TO DESCRIBE THE EVOLUTION OF THE DOMINANT TRAIT).

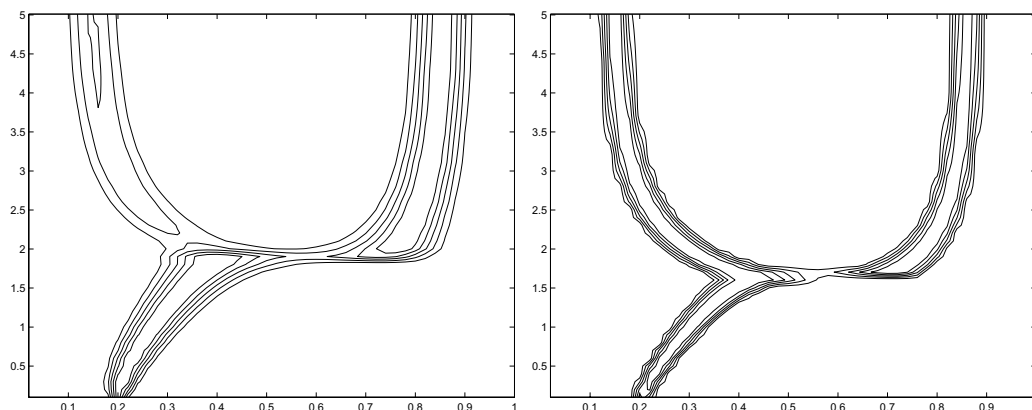


Figure 2.3: SAME AS FIGURE 2.1 FOR A TRAIT EVOLUTION WHICH EXHIBITS BRANCHING. THIS IS THE CASE OF A CHEMOSTAT WITH TWO NUTRIENTS.

Another and in some sense opposite example of possible evolution for a dimorphic population, is that it evolves toward a monomorphic situation because the ESS is attractive for both monomorphic and dimorphic populations. We borrow the Figure 2.4 from [51] where a system for adults and juveniles is treated.

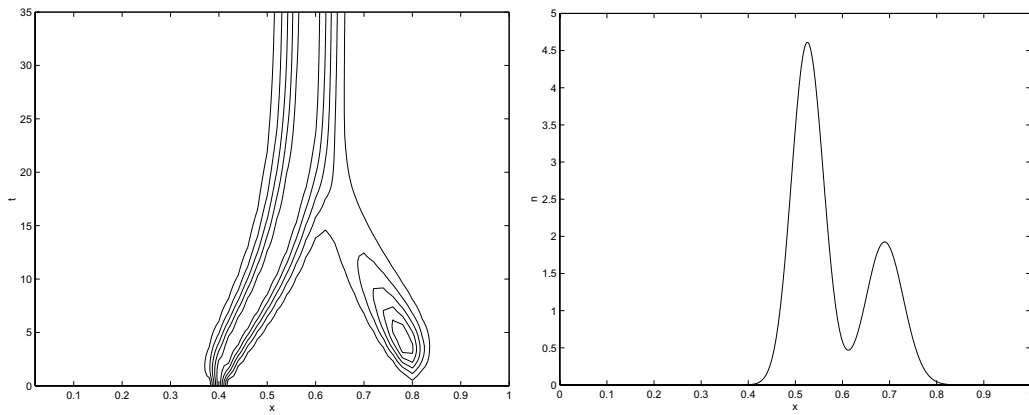


Figure 2.4: A DIMORPHIC POPULATION BECOMES MONOMORPHIC. TAKEN FROM [51] WHO CONSIDER A SYSTEM FOR ADULTS AND JUVENILES.